Pavan Nagesh

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Summary

Computer Science Engineering student with expertise in C++, Generative AI, Retrieval-Augmented Generation (RAG), and full-stack development. Hands-on experience building scalable applications through internships, freelance work, and hackathons. Strong team leader, communicator, and analytical problem solver committed to delivering high-quality software solutions.

Education

KLE Technological University — Hubli, Karnataka
Bachelor of Engineering in Computer Science — *2024 – 2027 (expected)*

KLE Smt C.I.M Polytechnic College — Hubli, Karnataka
Diploma in Computer Science — *2021 – 2024*

Skills

Languages: Python, C++, JavaScript, SQL, HTML, CSS
Frontend: React.js, Flutter
Backend & APIs: Node.js, FastAPI, REST APIs, Microservices, Authentication
AI & Machine Learning: Generative AI, LLMs, Retrieval-Augmented Generation (RAG), FAISS, XGBoost, Prompt Engineering, OCR
Blockchain: Solidity, Ethereum, Bitcoin, Web3, MetaMask, IPFS
Databases & Data Engineering: PostgreSQL, MongoDB, MySQL, MSSQL, Informatica Intelligent Data Management Cloud (IIDMC), ETL Pipelines, Power BI
Cloud & DevOps: Docker, Kubernetes, Git, Linux, CI/CD, Vercel

Experience

Freelance Full-Stack Developer – Klean Tech Systems — May 2026
Built and shipped SmartQuote, a production quotation and invoice automation platform for a retail business — *Hubli, India*

- Developed and deployed a SaaS platform used to manage 1,000+ products and generate quotations, invoices, and customer records for daily business operations.
- Implemented Excel-based bulk imports, automated GST calculations, and CI/CD on Vercel, reducing quotation preparation time by 80% and minimizing manual errors.

AI Research Intern – Traction Layer AI — Sep 2025 – Mar 2026
Built an Ambient AI platform for real-time clinical documentation and physician workflow automation — *Hubli, India*

- Developed an end-to-end AI pipeline with multilingual speech-to-text, speaker diarization, and LLM-based SOAP note generation to automate clinical documentation.
- Implemented a multi-agent RAG architecture to generate context-aware clinical notes grounded in hospital knowledge bases while ensuring accuracy and privacy compliance.

Project Intern – Infosys — Aug 2025 – Jan 2026
Built a decentralized slot booking system using Beckn Protocol — *Hubli, India*

- Architected a Beckn-compliant microservices system (Gateway, BAP, BPP, and Registry) for decentralized slot booking.
- Implemented asynchronous request-response workflows and orchestrated inter-service communication following Beckn Protocol specifications for decentralized service discovery and booking.

Projects

AI/ML Consensus-Aware Blockchain Simulator | *Python, Blockchain, Machine Learning*

- Designed and developed a novel Machine Learning-assisted blockchain simulator to analyze consensus performance by predicting throughput, latency, and energy consumption under varying transaction loads and mining difficulty.
- Conducted research on consensus optimization by integrating Machine Learning with blockchain performance analysis, resulting in a journal publication based on the proposed approach.

Multimodal Generative AI Copilot | Generative AI, RAG, Document Intelligence | *Python, FastAPI, LLMs, RAG*

- Developed a FastAPI-based multimodal AI copilot capable of understanding PDFs, images, audio, and text using Retrieval-Augmented Generation (RAG) for intelligent question answering and summarization.
- Implemented OCR and semantic document retrieval pipelines to improve factual accuracy, reduce hallucinations, and enable intelligent search across user-uploaded documents.

Achievements

Samved Hackathon 2026 — Third Prize, All India | *Top 3 of 500+ teams* | INR 50,000 Cash Award

AI-Driven Smart Water Distribution System | Flutter, Python, MQTT, ESP32, Machine Learning

- Developed an IoT-based water distribution system using ESP32, Arduino Mega, flow meters, and automated valves to monitor and control zone-wise water supply in real time via MQTT cloud communication.
- Built a Flutter dashboard and Python ML pipeline to forecast zone-wise demand, reducing supply imbalance by **20%** and improving allocation accuracy to **90%** across distribution zones.